

Exercise

Background: You are the chief health officer in the isolated city-state of SMDMopolis, which has its first ever documented case of **Virus X**. Virus X is an emerging infectious disease, and you have been tasked with projecting outbreak trends and identifying a policy response for the 10,000 residents of SMDMopolis.

Part 1: You review the limited literature from a handful of past outbreaks in other locations and find:

- People tend to feel better 14 days after developing symptoms.
 - In previous outbreaks, contact tracing studies have found that the average infected person infected between 3 and 10 other people.
1. Based on the information provided in the prompt, can you estimate best-case and worst-case effective contact rates (β) for Virus X?
 2. Combine your answer to part A with an SIR model to describe, under best- and worse-case scenarios:
 - a. The cumulative number of infections at 300 days (absent any intervention)
 - b. The likely size and timing of the epidemic peak (i.e., on which day is prevalence the highest, and how many new infections will that be?)

Part 2: In previous outbreaks, which tended to last 1-2 months, there were no documented instances of someone acquiring Virus X twice. However, a study has just been published which found that antibody levels wane substantially 3 months after recovery.

1. Update the model to incorporate waning immunity.
2. Incorporate this new data into your updated model.
 - a. Remember to keep your time scales (days, weeks, months) consistent.
3. How do your updates change your answers to exercise part 1? Consider again the best-case and worst-case scenarios.
 - a. The cumulative number of infections at 300 days (absent any intervention)
 - b. The likely size and timing of the epidemic peak
 - c. Describe your intuition for the directions and/or magnitude of changes.

Bonus questions (if time permits)

4. Does this update affect estimates of the basic reproductive number? How about the effective reproductive number?
 - a. Note: you do not have to recalculate any reproductive numbers – just answer yes/no and think about your reasoning.
5. You have also learned from SMDMopolis' chief demographer that last year, there were around 120 new births and 120 deaths documented in the population. Given the 300 day projections period you are considering, how important is it to incorporate demographics in

your model? Would your answer change if you were modeling over a longer timeframe (say, 10 years)?

- a. Note: we are not asking you to update your model here

Part 3: You have now learned that people tend to develop symptoms 5 days after being exposed to the virus. No evidence of subclinical transmission has been documented.

1. Update your model/code from Part 2 to incorporate latency.
2. Incorporate this new data into your updated model.
 - a. Remember to keep your time scales (days, weeks, months) consistent.
3. How do your updates change your answers to part 2? Consider again the best-case and worst-case scenarios.
 - a. The cumulative number of infections over 300 days (absent any intervention)
 - b. The likely size and timing of the epidemic peak
 - c. Describe your intuition for the directions and/or magnitude of changes.

Part 4: You are considering instituting two isolation policies for people with symptoms: 1) a policy with high quarantine coverage (daily rate = 0.5), and 2) a policy with low quarantine coverage (daily rate = 0.1).

1. Using the model from SEIRS_qrnt.R, compare cumulative numbers of infections between the two policies. Consider both best- and worst-case scenario of viral transmission.
2. Discuss the factors that might lead to the isolation policy with high quarantine coverage being less preferred compared to the policy with low quarantine coverage.

Part 4-ext: You are informed that the isolation policy with a higher coverage requires expansion of the current health care system and hence its implementation will cost more than the isolation policy with a low coverage. It is uncertain whether the additional cost is worth the health benefit. You are given the following additional information (see next page):

Parameters	Value
Total cost of low coverage isolation policy	\$5,000
Total cost of high coverage isolation policy	\$500,000
Disutility (Daily)	
Susceptible	0
Exposed	0
Infected	0.3
Recovered	0
Quarantined	0.3
Costs (Daily)	
Susceptible	\$0
Exposed	\$0
Infected	\$50
Recovered	\$0
Quarantined	\$100

1. Report the total cost and QALY (quality-adjusted life year) of each isolation policy, accrued over 300 days after the outbreak.
2. Report the incremental cost effectiveness ratio of the high coverage policy compared to the low coverage policy and make a decision at the willingness-to-pay threshold of \$20,000/QALY.